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Key Points:

- Well-preserved amino acid $\delta^{15}N$ tracks isotopic changes in upper ocean nutrient and the trophic structure of the western equatorial Pacific
- The grazing proportion exhibits distinct precessional variability and slightly lower values during Interglacial warmth than the Last Glacial
- Reduced grazing proportion may contribute to a decrease of oceanic carbon storage and an associated rise in atmospheric pCO_2

Supporting Information:

Supporting Information may be found in the online version of this article.

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Orbital Variability in Grazing Proportion: New Insights From Sedimentary Amino Acid $\delta^{15}N$ Records of the Western Equatorial Pacific

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Abstract The carbon sequestration capacity of the biological carbon pump (BCP) is determined by surface ocean carbon fixation and the transportation of fixed carbon to the ocean interior, which is closely linked to the trophic structure of planktonic ecosystem. However, previous biogeochemical reconstructions focused mostly on primary productivity, overlooking changes in community structure (e.g., ratio of primary producers to consumers). In this study, we investigate the late Quaternary history of planktonic trophic structure in the western equatorial Pacific by generating sedimentary amino acid $\delta^{15}N$ record of Site MD10-3340. Our results demonstrate well-preserved amino acid $\delta^{15}N$ signatures. Using the $\delta^{15}N$ gradients between trophic and source amino acids, we calculate the average sedimentary trophic position and design an idealized ecosystem model to quasi-quantitively estimate changes in grazing proportion (the proportion of primary production consumed by zooplanktons). Notably, lower trophic position and smaller grazing proportion point to a decline of zooplankton grazing during precession maxima, which is primarily attributed to enhanced nutrient availability in the western equatorial Pacific during El Niño-like mean-states. Moreover, average grazing proportion is slightly lower during Interglacial warmth than the last glacial period, implying a secondary control of sea level change. Furthermore, our ocean-biogeochemical simulations show that a halved grazing proportion could cause pronounced decrease in Pacific carbon storage and a remarkable increase in atmospheric pCO_2, due to a shallower remineralization depth. We thus highlight the necessity to incorporate varying marine grazing proportions into future proxy reconstructions and model simulations to facilitate a more comprehensive understanding of the BCP.

Plain Language Summary One of the most important ways for the ocean to absorb atmospheric CO_2 is the biological carbon pump, through which phytoplankton in the surface ocean convert CO_2 into organic carbon that sinks into the ocean interior. The trophic structure of the planktonic ecosystem, in terms of the proportion of phytoplankton-produced organic carbon consumed by zooplankton, is key for determining the ability of biological carbon pump, because it regulates the amount and types of settling organic carbon. In this study, we reconstruct the history of trophic structure in the western equatorial Pacific, by analyzing the nitrogen isotopes of amino acids preserved in the marine sediments. Our results show that trophic structure exhibits prominent precessional changes over the last 140,000 years. Specifically, less phytoplankton is grazed by zooplankton during the precession maxima, and grazing proportion is slightly lower in the interglacial warm periods than the Last Glacial. These changes have major impacts on the carbon cycle, since a substantially reduced grazing proportion would facilitate decreased carbon storage in the deep-ocean, thereby favoring an atmospheric CO_2 rise, according to our sensitivity experiments of model simulation.

1. Introduction

The ocean's capacity as a carbon sink is largely determined by the biological carbon pump (BCP), which transports carbon in organic forms from the surface ocean to depth (Ducklow et al., 2001). Without the BCP, the current atmospheric CO_2 level would be 200–300 ppm higher (Heinze et al., 2015). During ice ages, enhanced BCP is proposed to account for about one third (~30 ppm) of the glacial-interglacial reduction in atmospheric CO_2 (Kohfeld et al., 2005). Indeed, reconstructions based on the accumulation of organic debris show 50% higher organic carbon accumulation rate during the glacial maxima than during the interglacials (Cartapanis et al., 2016), with prominent excess glacial export production in the Subantarctic (Kohfeld et al., 2005). Nonetheless, iron fertilization that stimulates high-latitude productivity could explain only about 10 ppm of glacial-interglacial CO_2

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differences (Gottschalk et al., 2019), highlighting the need for a more comprehensive assessment of the BCP carbon sequestration capacity (Galbraith & Skinner, 2020).

The efficiency of the BCP, in terms of the fraction of primary production that escapes upper ocean recycling and is exported to the ocean interior (Buesseler & Boyd, 2009), depends largely on the remineralization of settling particulate organic carbon (POC) into dissolved inorganic carbon (DIC), including consumption by zooplankton and higher-level consumers, as well as degradation by microbes. Typically, less than 3% of net primary production reaches below 1,000 m (De La Rocha & Passow, 2007), where exported carbon can be sequestered over 1,000 years (Siegel et al., 2023). The sinking POC mainly consists of both phytoplankton-origin components (individual and loosely aggregated phytodetritus) and zooplankton-origin materials (carcasses, molts, mucous houses, and fecal pellets), whose vertical attenuation rate is highly sensitive to the organics' characteristics, like compositions and sinking velocities (Iversen, 2023). Thus, changes in the trophic structure may regulate the efficiency of the BCP by influencing the remineralization rate and depth of the settling POC (Bach et al., 2019).

In modern marine ecological studies, grazing proportion, representing the proportion of primary production consumed by zooplankton, is a widely used indicator of upper ocean trophic structure, with a contemporary global average of 0.66 (Steinberg & Landry, 2017). The grazing proportion is an important determinant of BCP, as it dictates the amount of POC fluxes leaving the surface ocean and the nature and degradability of the settling materials (Schmoker et al., 2013). However, assessing temporal variations in grazing proportion is challenging. While previous paleoceanographic studies have revealed large-scale transformation in sedimentary compositions over glacial cycles (e.g., Xu et al., 2025), they predominantly focus on the abundance and assemblages of specific planktonic flora (e.g., diatoms; Katsuki et al., 2003) or fauna (e.g., foraminifera or radiolarians; Herguera, 2000; Hernández-Almeida et al., 2017), lacking a quantitative constraint of the overall trophic structure.

The nitrogen isotopic composition ($\delta^{15}N = (^{15}N/^{14}N)_{sample}/(^{15}N/^{14}N)_{atmosphere} - 1$) of compound-specific amino acids ($\delta^{15}N_{AA}$) is a practical proxy that may simultaneously track changes in upper ocean trophic structure, monitor $\delta^{15}N$ of nutrients at the base of food web, and constrain the microbial degradation on organics (Ohkouchi et al., 2017). Amino acids (AAs) are the basic constituent of proteins, the basic compound of most life forms. Most amino acids can be grouped as either source amino acids (SRC-AA), whose $\delta^{15}N$ ($\delta^{15}N_{SRC}$) remains almost unchanged through food chains, or trophic amino acids (TR-AA), with their $\delta^{15}N$ ($\delta^{15}N_{TR}$) significantly enriched with each trophic step (McClelland & Montoya, 2002). As such, the difference between $\delta^{15}N_{TR}$ and $\delta^{15}N_{SRC}$ in specific organisms is commonly used to assess their trophic positions (Chikaraishi et al., 2007, 2009). Comparisons between sediment trap samples and surface sediments indicate well preserved $\delta^{15}N_{AA}$ signatures during settling (Batista et al., 2014; McCarthy et al., 2007). It is thus proposed that paired $\delta^{15}N_{SRC}$ and $\delta^{15}N_{TR}$ records in sediment cores can help to infer the overall picture of past changes in planktonic trophic structure (Batista et al., 2014).

This study focuses on the western equatorial Pacific (WEP), given the regional characteristics of low primary productivity but profound zooplankton contribution to total exported carbon out of the euphotic zone (Nowicki et al., 2022). Downcore $\delta^{15}N$ of sedimentary AAs at Site MD10-3340, spanning the last 140 ka, is generated to investigate the history of trophic structure as well as changes in original upper ocean nitrate isotopic signal. Specifically, we establish an idealized ecosystem model that quantitively estimates grazing proportion using $\delta^{15}N_{AA}$. Our results show distinct precessional variability in the average trophic position and calculated grazing proportion, as well as lower grazing proportion during the Interglacials of Marine Isotope Stage (MIS) 1 and 5 than the Last Glacial period of MIS 2–4. Finally, our ocean-biogeochemical model shows that a substantial reduction in grazing proportion could effectively reduce oceanic carbon storage, underscoring the tight link between planktonic trophic structure and the marine carbon cycle.

2. Materials and Methods

2.1. Study Site and Age Model

Site MD10-3340 (00°30.98′S, 128°43.47′E) is recovered during the MD181 MONOCIR cruise in 2010, from the Halmahera Sea in the WEP at a water depth of 1,094 m (Figure 1). Downcore sediments are assigned to a single lithologic unit composed primarily of homogeneous fossiliferous clay (Dang et al., 2015). The age model is established by radiocarbon dating on planktonic foraminifera *Globigerinoides ruber* and the $\delta^{18}O$ stratigraphy of benthic foraminifera *Uvigerina* spp. (Dang et al., 2015).

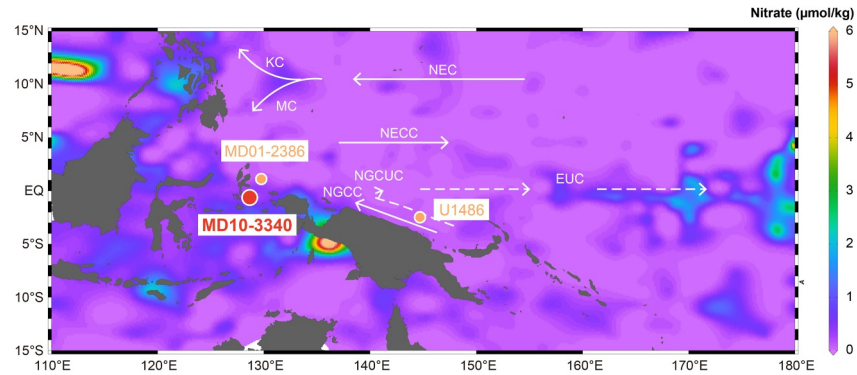


Figure 1. Map and site location of the western equatorial Pacific. Colored map shows the nitrate concentrations of the upper 20 m (data from World Ocean Altas 2018, Garcia et al., 2018). Solid arrows indicate the major surface currents: North Equatorial Current (NEC), Kuroshio Current (KC), Mindanao Current (MC), New Guinea Coastal Current (NGCC), North Equatorial Countercurrent (NECC). Dashed arrows indicate the subsurface currents: New Guinea Coastal Undercurrent (NGCUC) and Equatorial Undercurrent (EUC).

2.2. Organic Content Analyses

To eliminate potential distortion of terrestrial inorganic nitrogen, we analyze organic carbon (C_{org}%) and organic nitrogen content (N_{org}%) following the experimental procedures by Li and Jia (2011), using 72 samples from the top 17.2 m of Site MD10-3340. In general, carbonates and clay minerals are removed from sediments using hydrochloric acid (HCl) and hydrofluoric acid (HF). Organic remnants in acids are then collected by solid phase extraction and mixed with the residual sediments of acid treatments. Finally, about 2 mg of lyophilized pretreated samples are sealed in tin capsules and measured by elemental analyzer.

2.3. Amino Acids $\delta^{15}N$ Analyses

2.3.1. Sample Preparation

Approximately 2 g of grounded lyophilized sediments are used for amino acids $\delta^{15}N$ analyses, spiked with an internal AA standard (Isoleucine, Ile) with known concentration to monitor procedure losses and AA concentration. Pretreatment includes acid hydrolyzation, purification, and derivatization (modified from Li et al., 2019). Samples are hydrolyzed by 6 N HCl for 20 hr at 110°C. Hydrolysates are then passed through a 5-cm cation exchange resin column (Dowex 50WX8, 200–400 mesh) to isolate AA fractions. Within 6 weeks prior to isotopic measurements, derivatization is conducted on purified AAs using N-Acetyl Methyl Ester method (Corr et al., 2007a, 2007b), which esterifies carboxyl group by acid methanol and acetylates amino group with mixed solution of acetic anhydride, triethylamine and acetone. Final derivatized products are separated from salts with liquid-liquid extraction and stored in dichloromethane at −20°C.

2.3.2. Stable Isotope Analyses

Individual $\delta^{15}N_{AA}$ are measured using a Thermo Finnigan Trace 1310 gas chromatograph (GC) coupled to a Finnigan MAT 253 isotope ratio mass spectrometer (IRMS) through a combined combustion/reduction interface (Thermo Isolink II, 1000°C) and a liquid nitrogen cold trap. GC is fitted with an Agilent HP-5 column (60 m * 0.32 mm, 0.2 μm film thickness). Each measurement starts with a 30-s oxygen seed and a 1 μl sample injection in splitless mode (300°C). The oven temperature program starts from 80°C, increases by 8°C/min to 300°C, and ends with a 6-min final holding time. Derivatized external standard (USGS74, $\delta^{15}N = 30.17‰$) are run in a batch of three after every six sample injections. Final $\delta^{15}N_{AA}$ are calculated using a gradient correction to correct for instrument drift due to oxidation state of the reactor, followed by a second correction step based on the offset between known and measured $\delta^{15}N$ of the calibration standard.

Most samples are injected in duplicate to monitor the instrument performance, with the $\delta^{15}N$ standard deviation of injections (precision) varying between 0.5 and 1.3‰ for individual AAs (Table 1). To evaluate the potential isotopic influence of pretreatments, parallel samples are randomly taken at 10 depths, where the standard deviation of all injections at the same depth reflects the overall reproducibility, scattering from 0.5 to 1.5‰ for

Table 1
The $\delta^{15}N$, Mol Percentage, Instrumental Precision and Overall Reproducibility of Each Amino Acid

Group	Name	Average $\delta^{15}N_{AA}$	$\delta^{15}N_{AA}$ variation (standard deviation)	Instrumental precision[a]	Experimental reproducibility[b]	Average mol%	mol% variation (standard deviation)	Detection rate
SRC	Gly	12.2	1.6	0.9	1.0	6%	4%	86%
TR	Val	15.8	1.3	0.8	1.1	14%	5%	96%
SRC	Ser	14.1	1.8	1.3	1.3	2%	2%	61%
	Thr	5.3	2.2	1.3	1.5	2%	2%	61%
TR	Asp	13.9	1.4	0.5	0.5	20%	4%	98%
TR	Glu	15.8	1.5	0.8	1.0	9%	2%	100%
SRC	Phe	9.4	1.6	0.5	0.7	25%	5%	100%
SRC	Lys	8.1	2.1	0.6	1.0	14%	5%	96%
SRC	Tyr	9.4	1.2	1.0	1.3	8%	3%	90%

[a]average standard deviation of double injections. [b]average standard deviation of all injections of paralleled samples at the same depth.

individual AAs (Table 1). Additionally, the molar percentage (mol%) and concentration of individual AA is determined based on the peak amplitude of internal standard. Noted that AAs abundances could be affected by different choices of derivatization procedures, GC columns, and detection methods. All pretreatments and measurements in this study are conducted at Tongji University in Shanghai.

2.3.3. Parameter Definition and Statistical Analyses

The concentration and $\delta^{15}N$ are obtained for 4–9 individual amino acids in each sample, including (presented in the order of GC retention time): glycine (Gly), valine (Val), serine (Ser), threonine (Thr), asparagine (Asp), glutamic acid (Glu), phenylalanine (Phe), lysine (Lys), and tyrosine (Tyr). The concentration of all AAs in the sediments is measured by total hydrolyzable AAs (THAA) yields. Meanwhile, organic carbon-normalized THAA yields (OC-normalized THAA) shows the concentration of AAs per unit organic carbon.

The $\delta^{15}N$ of total hydrolyzable AAs ($\delta^{15}N_{THAA}$) is calculated as the molar-weighted mean $\delta^{15}N$ of all AAs detected in each sample (McCarthy et al., 2013):

$$\delta^{15}N_{THAA} = \sum(\delta^{15}N_{AA} * mol\%_{AA}) \quad (1)$$

Where $\delta^{15}N_{AA}$ and $mol\%_{AA}$ refer to the $\delta^{15}N$ and mol% of each AA. Error (standard deviation) of $\delta^{15}N_{THAA}$ is propagated from the reproducibility of individual $\delta^{15}N_{AA}$, averaging 1.0‰ across the entire sample set.

The trophic position (TP) describes the place of an organism in a food web, which is the energy transfer steps of the organism from the primary energy source. For example, primary producers like marine algae yield a TP of 1, while primary consumers like herbivorous zooplankton are designated TP of 2. In this study, average TP of all proteinaceous materials within the sample is determined based on the $\delta^{15}N$ difference between Glu and Phe (Chikaraishi et al., 2009):

$$TP = (\delta^{15}N_{Glu} - \delta^{15}N_{Phe} - \beta_{Glu-Phe})/TDF_{Glu-Phe} + 1 \quad (2)$$

Where $\beta_{Glu\text{-}Phe}$ (3.4‰) is the empirical difference between $\delta^{15}N_{Glu}$ and $\delta^{15}N_{Phe}$ in marine primary producers. The trophic discrimination factor ($TDF_{Glu\text{-}Phe}$ = 7.6‰) stands for an increase of $\delta^{15}N$ difference with each trophic transfer, derived from the empirical ^{15}N-enrichment factor of Glu (8‰) and Phe (0.4‰) along trophic level (Chikaraishi et al., 2009). The average propagated error of TP is 0.16, calculated based on the reproducibility of $\delta^{15}N_{Glu}$ and $\delta^{15}N_{Phe}$ measured in this study.

Decomposition of POM generally favors selective preservation of more stable compounds than the labile ones. The degradation index (DI) reflects the portion of more labile AAs in the sediments, with lower DI referring to a

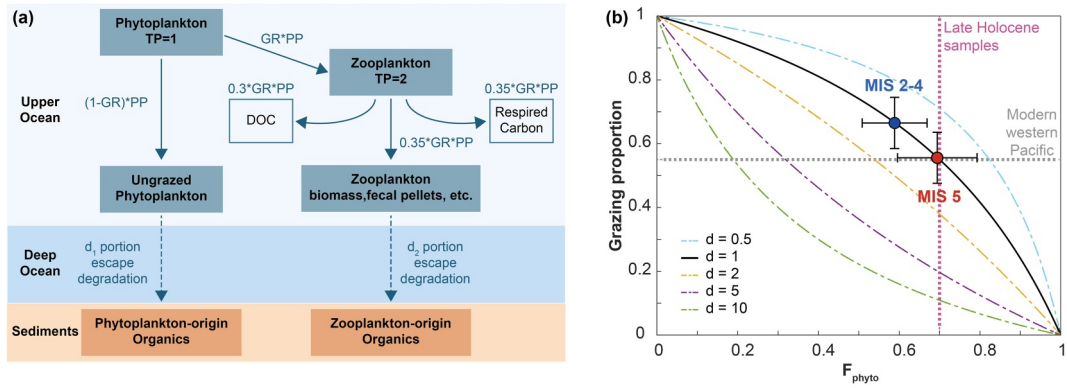


Figure 2. Calculating grazing proportion. (a) Schematic plot of the idealized ecological model, with an ocean box and a sediment box. Grazing proportion (GR) is the proportion of primary production (PP) being consumed by zooplankton, with the assumption that absorption efficiency is 70% and respiration rate is 50%. The proportions of organics that survives remineralization in the water column is d_1 for phytoplankton and d_2 for zooplankton, where d is the ratio of d_2/d_1. (b) Grazing proportion is a function of the fraction of phytoplankton-origin organics in the sediments (F_{phyto}) under various settings of d value. Average F_{phyto} and respective grazing proportion ($d = 1$) during MIS 2–4 and MIS 5 are shown by blue and red makers, along with 95% confidence interval. Dashed gray line represents the GR range of modern western Pacific (Steinberg & Landry, 2017). Dashed pink line marks the range of F_{phyto} of late Holocene samples at Site MD10-3340.

higher degree of diagenetic alteration. The DI is calculated based on the mol% ($mol\%_{AA}$) of all individual AA except for Lys:

$$DI = \sum[(mol\% - AVG_{AA})/STD_{AA}] * coef_{AA} \quad (3)$$

Where the average mol% (AVG_{AA}), mol% standard deviation (STD_{AA}), and factor coefficients ($coef_{AA}$) are given by Dauwe et al. (1999). Average propagated error of DI is 3.1.

Both animals and microbes can utilize POC to synthesize new protein of their own (i.e., heterotrophic proteinaceous resynthesis). Active animal and microbial heterotrophy would cause more scattered pattern of $\delta^{15}N$ in the TR-AA group, with microbial reworking exerting a more pronounced effect (McCarthy et al., 2004). We use the index $\sum V$ to indicate these heterotrophic resynthesis influences (McCarthy et al., 2007), calculated as the average deviation of individual $\delta^{15}N_{AA}$ in the trophic AA group to the mean $\delta^{15}N_{Tr}$:

$$\sum V = \left(\sum \left|\delta^{15}N_{AA} - \delta^{15}N_{Tr}\right|\right)/n \quad (4)$$

where n refers to the number of included TR-AA. Higher $\sum V$ refers to active heterotrophic proteinaceous resynthesis. As an internally normalized proxy, $\sum V$ is comparable among different studies. Average propagated error of $\sum V$ is 0.37.

2.4. Ecosystem Model and Biogeochemical Model

2.4.1. Ecosystem Model Settings

To quantitatively examine changes in the planktonic trophic structure, this study develops an idealized ecosystem model (Figure 2) to estimate changes in grazing proportion using AA-derived $\delta^{15}N$. The model is set up with three boxes (upper-ocean ecosystem, depositing POCs, and sedimentary AAs) and two types of organics (phyto- and zoo-planktons). In natural marine ecosystems, particularly in mid- to low-latitudes, the herbivorous zooplankton usually constitutes the majority of consumers (Nowicki et al., 2022). Our simplified model thus assumes only two trophic levels within the upper-ocean box, and corresponding settling phytoplankton-origin organics (TP = 1) and zooplankton-origin organics (TP = 2). In the sediment box, the fraction of phytoplankton-origin organics (F_{phyto}) is determined by $\delta^{15}N_{Glu}$, $\delta^{15}N_{Phe}$, and their empirical $\delta^{15}N$ differences in phytoplankton and zooplankton (detailed calculation in Text S1 in Supporting Information S1):

$$F_{phyto} = \left(11.4 - \delta^{15}N_{Glu} + \delta^{15}N_{Phe}\right)/8 \quad (5)$$

The propagated standard error of F_{phyto} is calculated based on the reproducibility of $\delta^{15}N_{Glu}$ and $\delta^{15}N_{Phe}$ in this study, using monte carlo method (sampling size = 10^6).

The next step involves quantifying the grazing proportion within the upper ocean box, which first considers the flow and conversion of primary production therein. Following the framework of most bioenergetic calculations, we assume a constant absorption efficiency of 70% (Steinberg & Landry, 2017), which is the proportion of grazed organic content absorbed by zooplankton digestive system (utilized for growth), whereas the unabsorbed organics are released back into the water column as dissolved organic matter (Figure 2a). Of the absorbed organics, approximately 50% is respired by zooplankton (Steinberg & Landry, 2017), leaving the other half depositing to deeper waters. Our model then represents the remineralization of settling organic matters in the deep ocean box by incorporating different proportions of surviving organics for phytoplankton and zooplankton. Following Equation 5, the grazing proportion (GR) can be expressed as follows (detailed in Text S1 in Supporting Information S1):

$$GR = (1 - F_{phyto})/(1 - F_{phyto} + 0.35 * F_{phyto} * \mathbf{d}) \quad (6)$$

Here, $\mathbf{d}_1$ and $\mathbf{d}_2$ represent the proportions of surviving organic matter derived from phytoplankton and zooplankton, respectively. For simplicity, we define $\mathbf{d} = \mathbf{d}_2/\mathbf{d}_1$. The choice of $\mathbf{d}$ value will be discussed in details in Section 4.3.2. Similar to F_{phyto}, propagated standard error of $\mathbf{d}$ is calculated using monte carlo method.

2.4.2. Biogeochemical Model and Experimentation

To investigate the impact of varying grazing proportion on oceanic carbon storage, this study employs a cutting-edge ocean biogeochemical model, specifically a coupled version of the Massachusetts Institute of Technology ocean general circulation model (MITgcm) (Campin et al., 2020) integrated with the Regulated Ecosystem Model 2 (REcoM2) (Hauck et al., 2013), following the work by Du and Tian (2025). The physical configuration of the MITgcm includes a dynamic-thermodynamic sea ice model (Losch et al., 2010). The MITgcm has a longitudinal resolution of ~2° and a latitudinal resolution of 0.38–2°, along with 30 vertical layers unevenly distributed across 5,500 m water depth. The biogeochemical model REcoM2 effectively simulates the cycles of carbon, nitrogen, iron, and silicon in the ocean, with a dynamic and globally uniform atmospheric pCO_2 box (Du et al., 2022), so that pCO_2 change is prognostic in the simulations. Specific model configurations have been used to investigate carbon cycle for present day (e.g., Pagnone et al., 2019) and for the past (see detailed description in Text S2 in Supporting Information S1).

We incorporate two experiments. One control run which corresponds the Pre-Industrial configuration, representing idealized interglacial conditions. Another run, serve as a sensitivity test, uses a lower grazing proportion of 0.2 (halved from the original value), with all other setups being unperturbed. For each experiment, we initialize the model from present-day observations and run the model for 5,000 model years. Detailed description of the model, initialization and forcing fields can be found from the previous study and the references therein (Du & Tian, 2025).

3. Results

At Site MD10-3340, the C_{org}% and N_{org}% vary between 0.53%–1.44% and 0.04%–0.13% for the last 140 ka (Figure 3a, Figure S1a in Supporting Information S1), both presenting lower values during MIS 2–4 and higher contents during MIS 1 and 5. There is a strong correlation between C_{org}% and N_{org}% (prewhitened correlation $r^2 = 0.83$, $p < 0.05$) with a near-zero intercept. This suggests that inorganic nitrogen is effectively removed from the sediments, and all nitrogen analyzed in our following measurements are bounded with organic carbon (i.e., in organic forms). Moreover, a typical C/N_{org} ratio of 11 indicates a predominance of marine-sourced organic matter (Figure S1a in Supporting Information S1).

The THAA content ranges from 0.05 to 0.17 mg/g in sediments, showing a clear downcore decreasing trend and two major jumps across the last two deglaciations (transitions of MIS 6 to 5 and MIS 2 to 1; Figure 3b). Downcore OC-normalized THAA varies over 0.4–2.3 mg/100 mg OC (Figure 3b). Phe and Asp are the most abundant AAs,

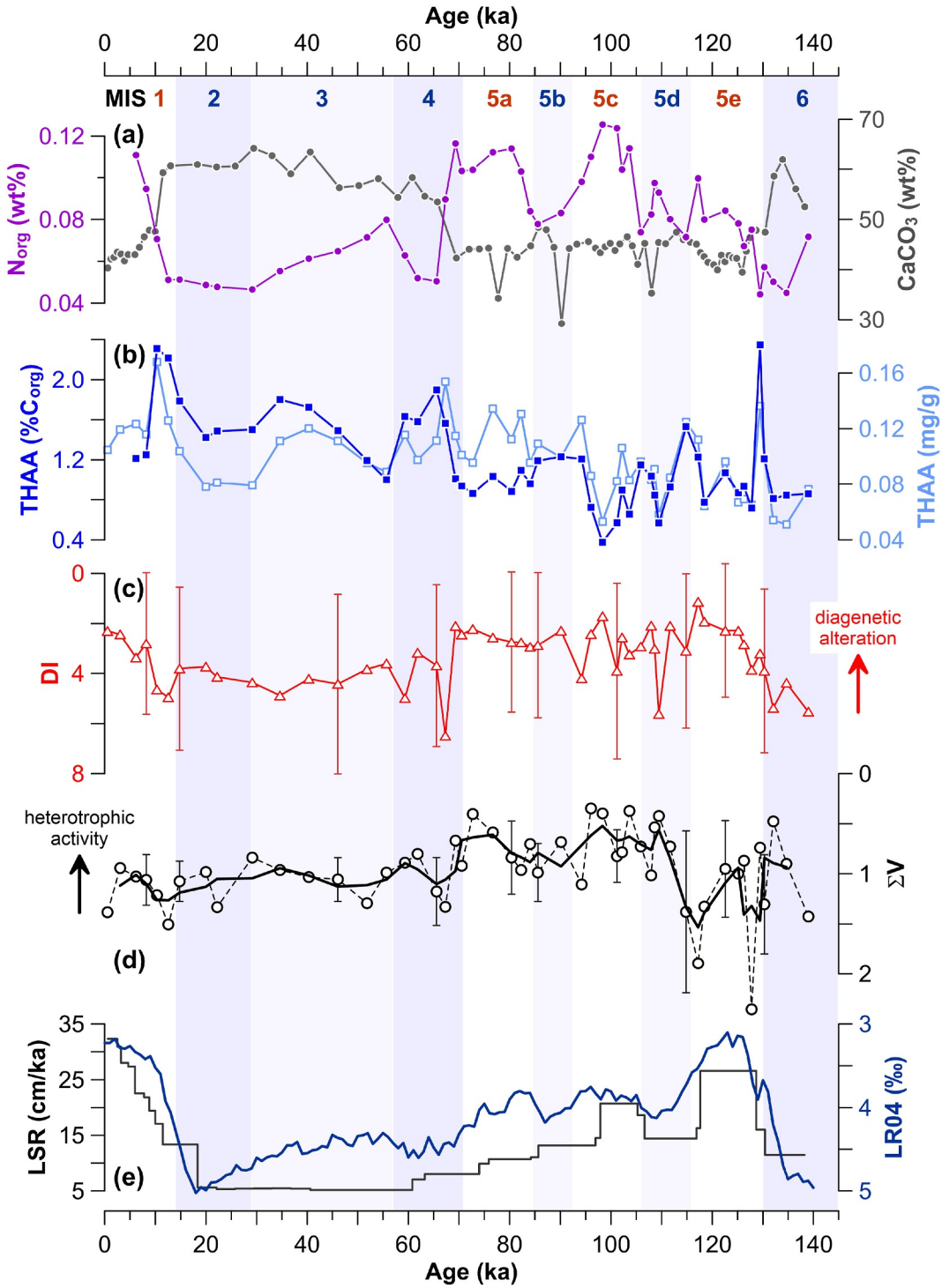


Figure 3. Concentration and diagenetic alteration indexes of amino acids at Site MD10-3340. (a) Organic nitrogen (N_{org}%, purple) and calcium carbonate ($CaCO_3$%, gray) concentration. (b) Total hydrolyzable AAs (THAA) content (light blue) and organic carbon normalized THAA yields (dark blue). (c) degradation index (DI). (d) Total heterotrophic AA resynthesis index, with the solid line showing LOESS smoothing data (smoothing factor = 0.1) ($\sum$V). (e) Linear sedimentation rate (black) and LR04 benthic stack (blue; Lisiecki & Raymo, 2005). Error bars in c and d show propagated standard deviation (2σ). Numbers on top of the figure indicate Marine Isotope Stages.

with average molar percentages of 25% and 20%, respectively (Table 1). Meanwhile, Ser and Thr were only detected in 61% of the samples, showing lowest molar percentage around 2%. Such low concentration results in relatively poor precision and reproducibility of $\delta^{15}N$ analyses on Ser and Thr (Table 1).

The DI generally fluctuates between 1 and 6, and is generally higher during MIS 1 and MIS 5 (Figure 3c). Noted that the interpretation of the DI data requires caution due to its relatively large error bars. The $\sum V$ mostly falls between 0.4 and 1.5‰ (Figure 3d). Although the average values of $\sum V$ do not change much from MIS 1 to MIS 6, its standard deviations increase in older sediments (0.2 for MIS 1 and MIS 2–4, 0.4–0.5 for MIS 5 and MIS 6).

In the SRC-AA group, Phe, Lys, and Tyr demonstrate lower $\delta^{15}N$ values (averaging between 8.1 and 9.4‰, Figure 4a) than Gly and Ser (averaging at 12.2 and 14.1‰, Figure S1b in Supporting Information S1). The $\delta^{15}N$ values of three TR-AA (Val, Asp, and Glu) are distinctly higher than those of the SRC group (Figure 4b), averaging from 13.9 to 15.8‰. Thr, a unique AA outside of either the SRC or TR group, shows the most depleted average $\delta^{15}N$ (average 5.3‰; Figure S1c in Supporting Information S1) and the largest standard deviation (2.2‰). The $\delta^{15}N$ difference between Glu and Phe ranges 3.6–8.3‰, resulting in TP ranging 1.0–1.6 (Figure 4c). The $\delta^{15}N_{THAA}$ varies about an average value of 11.7‰ (between 8.5 and 14.2‰) over the last 140 ka (Figure 4d).

4. Discussion

4.1. Evaluating the Imprint of Post-Deposition Alterations

The application of $\delta^{15}N$ from various sedimentary archives has long been confronted with the potential interference of diagenesis, which may induce compositional and isotopic modifications of organic matter (Altabet & Francois, 1994; Dauwe et al., 1999), especially when low sedimentation rates lead to longer seafloor exposure time of organics (Altabet et al., 1999; Thunell et al., 2004). Specifically, organic matter enriched in nitrogen (e.g., AAs and proteins) are more easily degraded by microbes (Lehmann et al., 2002). To assess the potential impact of early diagenesis and microbial degradations within sediment core, we perform the following analyses on both the compositional yields and $\delta^{15}N$ signatures of AA.

At Site MD10-3340, THAA content generally decreases downcore, implying slightly increased organic remineralization with burial depth (Figure 3b). Meanwhile, the OC-normalized THAA yields enable a direct comparison of AA preservation (relative to other organics) between sediments from the oligotrophic WEP and the productive eastern Pacific. At Site MD10-3340, downcore OC-normalized THAA yields are less than one-tenth of those in the surface sediments of the Santa Barbara Basin (40–10 mg/100 mg OC in the upper 40cm sediments; Batista et al., 2014), indicating intensive THAA loss at the seafloor. The OC-normalized THAA yields align with the depth profile of calcium carbonate concentration ($CaCO_3$%) and DI (Figures 3a–3c), typically below 1 mg/100 mg OC during MIS 5 (with ~45% $CaCO_3$ content and lower DI, thus stronger degradation), but rising above 1.5 mg/100 mg OC during MIS 2–4 (corresponding to ~60% $CaCO_3$ content and higher DI, thus better preservation). Similarly, covariation between DI values and the OC-normalized THAA downcore is also found in the Santa Barbara Basin (Batista et al., 2014), implying preferential AA degradation (relative to other organics) dominated by sedimentary composition. Nevertheless, none of the AA-derived $\delta^{15}N$ (Figure 4) shares similar patterns with either THAA yields, DI (Figures S2a–S2c in Supporting Information S1 for raw correlation; Figures S3a–S3c in Supporting Information S1 for prewhitened correlation), or linear sedimentation rate (Figure 3e), suggesting no interdependence between isotopic signal and the content or composition of AA.

We also assess the potential microbial alteration on $\delta^{15}N_{AA}$ using the total resynthesis index $\sum V$, which is a pure isotopically derived parameter irrespective of AA compositions. At Site MD10-3340, the oscillations of $\sum V$ appear to be slightly larger in older sediments (Figure 3d). Overall, $\sum V$ values (0.4 ~ 2.4) at this site are close to that of suspended and sinking organic particles in the high-nutrient-low-chlorophyl region of the northeast Pacific (1.2 ~ 2.7; Wojtal et al., 2023), implying similar extent of microbial activities in these low-productivity environments. Notably, the $\sum V$ values of MD10-3340 are even slightly lower than previously reported surface sediment values in the high-productivity Santa Barbara Basin and the eastern tropical Pacific (1.4 ~ 3.7; Batista et al., 2014; Sauthoff, 2016), where high productivity and high sedimentation rate are thought to facilitate better preservation of organics. This argues against severe microbial reworking on the WEP sedimentary AA. It is thus proposed that the $\delta^{15}N$ is well preserved in the sedimentary AA, faithfully tracking the isotopic signature of organic materials (protein and AA) sinking from the upper ocean, in spite of their clear degradation in core MD01-

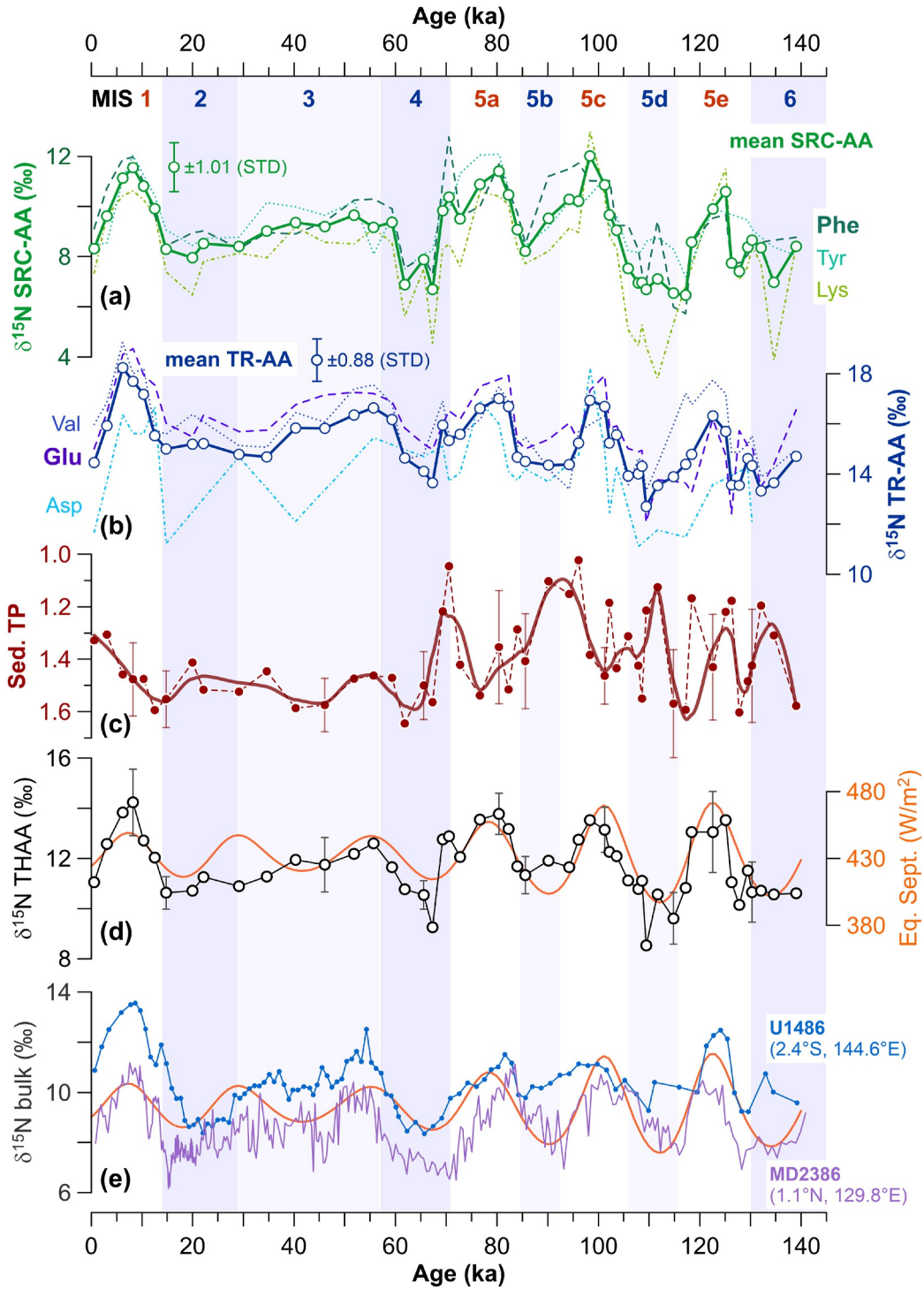


Figure 4. $\delta^{15}N_{AA}$ and average sedimentary trophic position at Site MD10-3340. (a) $\delta^{15}N$ of three representative source amino acids (SRC-AA) and their average value (solid green line). (b) $\delta^{15}N$ of three trophic amino acids (TR-AA) and their average value (solid blue line). (c) Average trophic position (TP) of all proteinaceous materials within the sediments, calculated using the $\delta^{15}N$ of Glu and Phe. The solid red line shows Loess smoothing data (smoothing factor = 0.1). (d) $\delta^{15}N$ of total hydrolyzable AAs ($\delta^{15}N_{THAA}$, in solid black line). (e) The western equatorial Pacific bulk sediment $\delta^{15}N$ ($\delta^{15}N_{bulk}$) records at Site MD01-2386 (purple; Jia & Li, 2011) and U1486 (blue; Lambert et al., 2022). The orange lines superimposed on d and e represent September insolation at the equator. Error bars in c and d show propagated standard deviation (2σ). Numbers on top of the figure indicate Marine Isotope Stages.

3340. This extends the utility of sedimentary $\delta^{15}N_{AA}$ proxy from high-productivity eastern Pacific to a broader region with lower production and more oxic bottom waters.

4.2. Tracking the Signatures of Original Nitrate $\delta^{15}N$

AA-derived $\delta^{15}N$ has been proposed to be a supplement of $\delta^{15}N_{bulk}$ in paleoceanographic applications, since $\delta^{15}N_{bulk}$ is thought to be prone to diagenetic alteration and contamination from terrigenous inorganic ammonium (Schubert & Calvert, 2001). At Site MD10-3340, paralleled variations spanning the last 140 ka are observed in $\delta^{15}N$ of nine AAs (Figures 4a and 4b; Figures S1b and S1c in Supporting Information S1). The first principal component (PC1; Figure S1e in Supporting Information S1) of all $\delta^{15}N_{AA}$ accounts for ~60% of the total data variance, with each individual $\delta^{15}N_{AA}$ showing positive loading of 0.64–0.87. Strong positive correlations between $\delta^{15}N_{THAA}$ and PC1 for both raw data ($r^2 = 0.8$, $p < 0.05$, Figure S2h in Supporting Information S1) and prewhitened data ($r^2 = 0.56$, $p < 0.05$, Figure S3h in Supporting Information S1) confirm that $\delta^{15}N_{THAA}$ reflected an overall pattern of all proteinaceous materials.

Earlier studies on individual marine organisms inferred that $\delta^{15}N_{Phe}$ presents minimal fractionation along the food web, thus represents a direct proxy of source nitrate $\delta^{15}N$ (Chikaraishi et al., 2007, 2009). Meanwhile, later work observed an inverse relationship between $\delta^{15}N_{Phe}$ and TP in a "closed system" (Batista et al., 2014). Interestingly, we observe a weak correlation between the raw data of $\delta^{15}N_{Phe}$ and TP ($r^2 = 0.18$, $p < 0.05$, Figure S2f in Supporting Information S1), but a much better prewhitened correlation ($r^2 = 0.45$, $p < 0.05$, Figure S3f in Supporting Information S1) after fitting an Autoregressive Moving Average (ARMA) model to each series and removing the predictable components. This suggests a short-term interdependence of $\delta^{15}N_{Phe}$ and TP, but argues against a steady-state isotope mass balance assuming a fixed $\delta^{15}N$ of the proteinaceous pool on orbital or longer timescales. Instead, the strong correlations between $\delta^{15}N_{Phe}$ and $\delta^{15}N_{THAA}$ ($r^2 = 0.62$ for raw correlation, Figure S2g in Supporting Information S1; $r^2 = 0.57$ for prewhitened correlation, Figure S3g in Supporting Information S1) indicate an overall dominance of source nutrient $\delta^{15}N$ over isotopic fractionations induced by trophic transfer. Thus, both $\delta^{15}N_{Phe}$ and $\delta^{15}N_{THAA}$ mainly reflect the $\delta^{15}N$ changes (despite of different absolute values) of nutrient at the base of the food web. Specifically, in the WEP, where surface nutrient is completely consumed, nutrient $\delta^{15}N$ at the base of the food web generally equals the $\delta^{15}N$ of source nitrate mainly originating from the subsurface (Rafter et al., 2012).

Notably, there are comparable fluctuations among $\delta^{15}N$ records in the WEP. During the last 140 ka, changes in $\delta^{15}N_{THAA}$ and $\delta^{15}N_{Phe}$ at Site MD10-3340 are highly consistent with the $\delta^{15}N_{bulk}$ record of MD01-2386 near Halmahera Island (Jia & Li, 2011) and $\delta^{15}N_{bulk}$ at Site U1486 northeast off Papua New Guinea (Lambert et al., 2022) (Figure 4e). Besides the clear glacial-cycle variabilities, downcore $\delta^{15}N$ at these three WEP sites are dominated by precession, manifesting a synchronization with September insolation at the Equator (Figures 4d and 4e). The distinct and coherent precession cycles (23.3 ka) and secondary obliquity cycles (40 ka; Figures S4a and S4b in Supporting Information S1) have been attributed to local nitrogen fixation (Jin et al., 2024) and high-latitude controls (Lambert et al., 2022).

Here, we focus on comparing different $\delta^{15}N$ proxies within the same region, rather than delving into the mechanism behind orbital cycles in upper WEP nitrate $\delta^{15}N$, which is beyond the scope of this study. The consistency across the three sites precludes any significant fractionation from site-specific diagenesis. Furthermore, since AA-derived $\delta^{15}N$ exclusively reflects marine-sourced organics (Keil & Fogel, 2001), the remarkable similarities between AA-derived $\delta^{15}N$ and $\delta^{15}N_{bulk}$ records indicate minimal terrigenous isotopic contamination. Hence, $\delta^{15}N_{bulk}$ is suggested to be a robust proxy of upper ocean nitrate $\delta^{15}N$ in the WEP, despite the low productivity, oxic depositional environments and substantial terrestrial inputs in WEP.

4.3. Assessing the Variations in Marine Grazing Proportion

4.3.1. Varying Average Trophic Position

Trophic fractionation may be also imprinted in the isotopic signatures of AA when they channel through the food web. Calculated based on the gradient between $\delta^{15}N_{Glu}$ and $\delta^{15}N_{Phe}$, average sedimentary TP ranges 1.0–1.6 over the last 140 ka (Figure 4c), indicating a mixed planktonic source material with varying contributions of primary producers (TP = 1) and consumers (TP > 1). For the late Holocene, estimated TP of Site MD10-3340 is ~1.3, close to that of trap sediments in the central Equatorial Pacific at a similar depth (~1,000 m) (McCarthy

et al., 2013). Large TP variability at our site is comparable to those of eastern tropical Pacific over the Holocene (Sauthoff, 2016). The TP generally maintains at 1.4–1.6 during MIS 2–4, but shows lower values and larger-amplitude oscillations (1.0–1.6) during MIS 5. An independent sample *t*-test indicates a marginally significant difference between the average TP during MIS 2–4 and MIS 5 ($t = 2.04$, $p = 0.048$). Notably, dynamics of pelagic food web are in concert with changes in precession index, with higher sedimentary TP, thus larger proportion of consumers (e.g., zooplankton) and less primary producers (e.g., phytoplankton), during precession minima than precession maxima.

Moreover, the $\delta^{15}N$ of Thr shows the opposite trend to the trophic AAs (e.g., $\delta^{15}N_{Glu}$) at Site MD10-3340. Smaller gradient between $\delta^{15}N_{Phe}$ and $\delta^{15}N_{Thr}$ ($\Delta\delta^{15}N_{Phe\text{-}Thr}$) is observed during MIS 5 (Figure S1d in Supporting Information S1). This agrees with modern ecological studies with phytoplankton, zooplankton and marine mammals, which suggest that $\delta^{15}N_{Thr}$ becomes progressively depleted with increasing TP (Germain et al., 2013). The inverse trend of TP and $\delta^{15}N_{Thr}$ is also evident in organic particles in the water column. While TP gradually increase from ∼0.9 in the surface ocean to ∼2.3 at the mesopelagic depth due to cumulative contribution of heterotrophic activity, $\delta^{15}N_{Thr}$ exhibits a drastic downward decrease, especially for larger particles (Wojtal et al., 2023).

4.3.2. Large-Amplitude Changes in Marine Grazing Proportion

Conventionally, paleoenvironmental studies invoke primary/export productivity to indicate the overall downward POC flux, for the subsequent analyses of marine BCP efficiency. This assumption is reasonable since the phytoplankton production forges the basis of the upper-ocean food web, but it overlooks the biomass and effects of zooplankton and higher-level consumers. For example, observations over the last decades suggest increasing chlorophyl in the surface eastern tropical north Pacific accompanied by declining zooplankton biomass (Lavaniegos & Ohman, 2003, 2007), which is recorded in the declining trend of sedimentary TP (from 1.5 to <1, Batista et al., 2014). Our records of average TP, with large range over the last 140 ka, implies that the ratio of consumers/producers in the upper-ocean ecosystem may vary profoundly.

To quantitatively examine such changes, we develop an idealized ecosystem model to reconstruct past changes in grazing proportion (*Methods*; Figure 2a). The first step is determining the fraction of phytoplankton-origin organics in the sediments (F_{phyto}) using $\delta^{15}N_{Glu}$ and $\delta^{15}N_{Phe}$ (Equation 5 *in Methods*). Calculated F_{phyto} at site MD10-3340 shows late Holocene (0–4 ka) value of ∼0.70 (when TP = 1.3), varying from ∼0.56 during MIS 2–4, ∼0.67 during MIS 5, and ∼0.64 at the end of MIS 6 (Figure 4b). This calculation seems to underestimate the contribution of zooplankton, which accounts for >70% of the contemporary carbon exported out of the WEP euphotic zone as estimated by some global ecological models (Nowicki et al., 2022). Nonetheless, endmember analysis using Phe-normalized $\delta^{15}N_{Thr}$ and $\delta^{15}N_{Ala}$ values of in situ filtered particles in the northeast Pacific gives estimates closer to our results (Doherty et al., 2021). For detrital organics with TP at ∼1.3, fecal pellet contribution is <0.4, while phytoplankton/phytodetritus and microbially degraded organic matters comprise the remainder (Wojtal et al., 2023).

Next, grazing proportion is calculated with F_{phyto} and the degradation coefficient $\mathbf{d}$ (Equation 6 *in Methods*). $\mathbf{d}$ is defined as $\mathbf{d}_2/\mathbf{d}_1$, where $\mathbf{d}_1$ and $\mathbf{d}_2$ represent respective surviving proportions of phytoplankton and zooplankton during settling (see Text S1 in Supporting Information S1). To determine a reasonable value of $\mathbf{d}$, we compared reported grazing proportion in the modern tropical Pacific (0.55) (Calbet & Landry, 2004; Schmoker et al., 2013) with the late Holocene F_{phyto} at site MD10-3340 (0.70), whose cross-plot aligns well with the GR estimation line for $\mathbf{d} = 1$ (Figure 2b). Notably, the grazing proportion exhibits a dominant variability on precession band, with lower grazing proportion coinciding with precession maxima (Figure 5c). Analogous estimations with $\mathbf{d} = 0.5$ or $\mathbf{d} = 2$ yield comparable trends of GR (Figure S5 in Supporting Information S1). Variations in filtered grazing proportion are very similar to the variation in the amplitude of precession index (Figure 5d), showing pronounced variabilities during MIS 5, but remains relatively stable over MIS 2–4. Spectral analysis reveals a distinct precessional periodicity around 18 ka for the grazing proportion (Figure S4c in Supporting Information S1), despite the relatively short coverage and coarse resolution of our records.

On the very first order, changes in the trophic structure are closely related to nutrient availability in the upper WEP, which is intrinsically modulated by upper ocean dynamics. The precession-driven grazing proportion change can be attributed to the upward entrainment of deeper nutrients and associated primary production in the WEP. Such a precession-dominated variability has been found in the Eastern Equatorial Pacific (EEP) $\delta^{15}N$

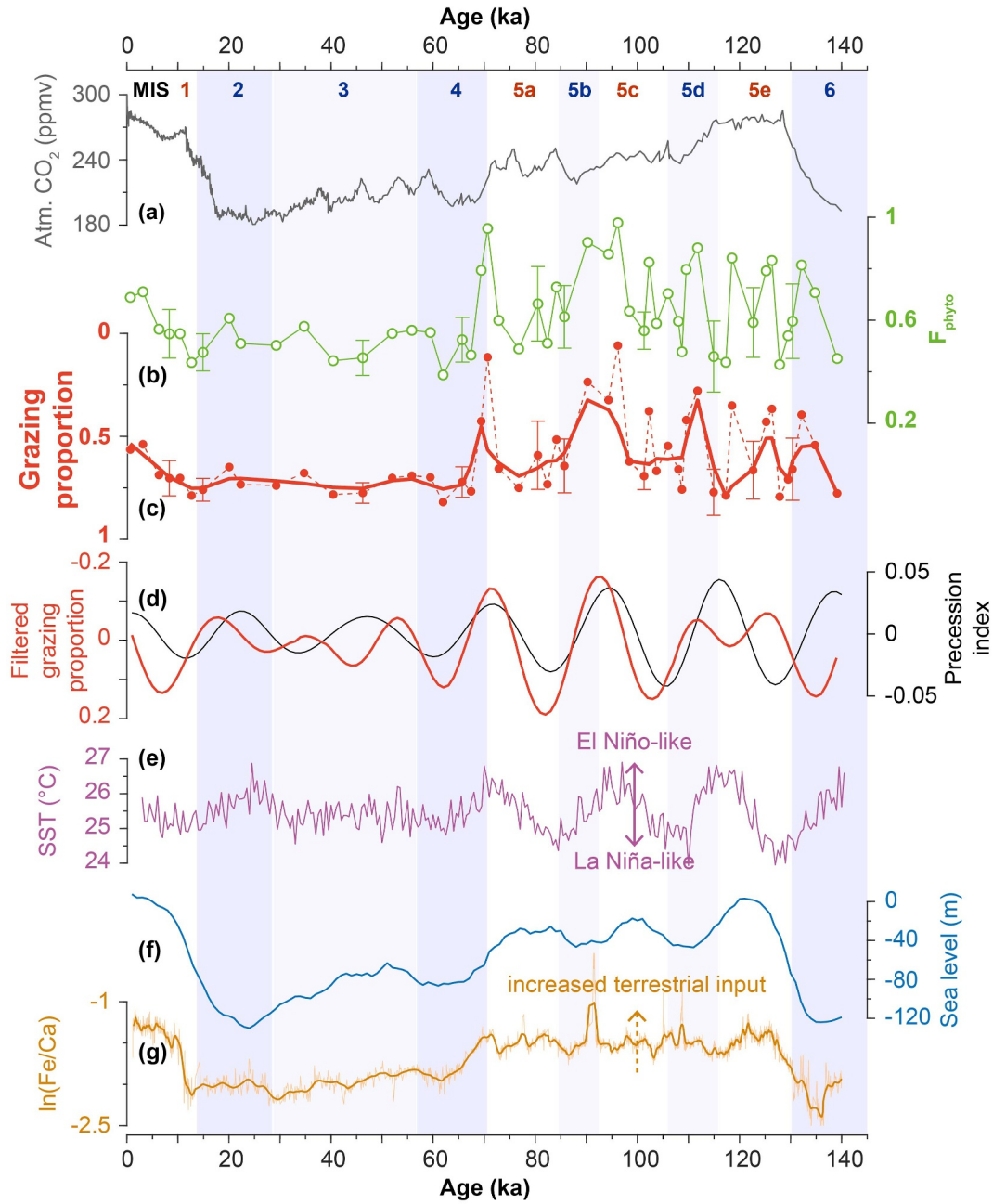


Figure 5. Reconstructing the western equatorial Pacific ecological structure. (a) Composite atmospheric CO_2 concentration from Antarctic ice cores (Bereiter et al., 2015). (b) The fraction of phytoplankton-origin organics in Site MD10-3340 sediments (F_{phyto}). (c) Grazing proportion calculated with F_{phyto} under settings of d = 1. (d) Precession index (black) and filtered grazing proportion on precession band. (e) Modeled Niño 3 winter sea surface temperature (SST). Smaller Niño 3 refers to La Niña-like mean state (Zhang et al., 2021). (f) Global composite sea level stack (Spratt & Lisiecki, 2016). (g) Normalized sedimentary iron contents (ln(Fe/Ca)) at Site MD10-3340 (Dang, Wu, et al., 2020). Error bars in b and c show propagated standard deviation. Numbers on top of the figure indicate Marine Isotope Stages.

records and interpreted to be driven by an upper-ocean physical processes, that is, upwelling, that regulates the nutrient utilization (Rafter & Charles, 2012). From the point of view of thermocline connection across WEP and EEP (Jian et al., 2022, 2023) under precession-driven ENSO-like mean-state fluctuations (Clement et al., 1999), we further propose that during precession maxima, the equatorial Pacific is dominated by El Niño-like mean states, which would enhance vertical nutrient supply to the WEP surface due to a reduced upper ocean

stratification (Figure 5e; Zhang et al., 2021). While phytoplankton productivity responds directly and rapidly to changes in nutrient supply, there could be a decline of relative importance of zooplankton grazing with increasing primary production (Calbet, 2001).

Despite the clear precessional changes, a difference in the average values of the grazing proportion could be recognized between the Interglacial warm intervals and Glacial period. With $\mathbf{d} = 1$, the average grazing proportion increases from 0.55 ± 0.20 during MIS 5 and Late Holocene to 0.67 ± 0.18 during MIS 2–4. These glacial-interglacial differences could be related to the modulation of sea level. During periods of low sea level (Figure 5f), the development of the ascending branch of the Walker circulation over the Maritime Continent would be suppressed (DiNezio & Tierney, 2013), decreasing precipitation and associated terrestrial nutrients input from these WEP islands (Figure 5g; Dang, Wu, et al., 2020). Thereby the growth of phytoplankton would be limited, and resulting in generally higher grazing proportion during MIS 2–4.

4.4. Testing the Outcome of Varying Grazing Proportion

Upper ocean trophic structure potentially affects the carbon sequestration efficiency of biological bump in two opposite ways. The canonical understanding associates a lower grazing proportion with increased downward POC flux and higher BCP efficiency, due to a suppressed POC recycling in the uppermost waters (Mayzaud & Pakhomov, 2014). However, the sequestration efficiency of exported carbon also depends on how remineralization is distributed with depth. Compared to the low sinking velocities of phytoplankton individuals (<1–10 m/day) and loosely aggregated phytodetritus (100–150 m/day; Turner, 2015), denser zooplankton-origin organics are rapidly exported to depth (up to 800 m/day; Robison et al., 2005) and may be more refractory (Gleiber et al., 2012; Liszka et al., 2019). Thus, despite fast nutrient recycling in the euphotic zone, a higher grazing proportion may bring POC remineralization to deeper depths of mesopelagic region (~100–1,000 m), which potentially lengthens the carbon sequestration time (Boyd et al., 2019; DeVries et al., 2012; Siegel et al., 2023).

To evaluate the integrated effects of these two opposite mechanisms, the broad direction of the outcome of changing grazing proportion on the carbon cycling is tested using a biogeochemical model coupled to an oceanic general circulation model (Du & Tian, 2025). Take typical interglacial warm period as an example, the initial state of the model applies a grazing proportion (grazing efficiency parameter) of 0.4 for the global ocean (Text S2 in Supporting Information S1). We then reduced the global grazing proportion by 50% to 0.2 in the sensitivity experiment, with the other boundary conditions held the same (*Methods*). In the model, the reduced parameter primarily suppressed the carbon and hence nutrient inventory in the zooplankton, with the most prominent reduction in the tropical-subtropical oceans (Figure S6 in Supporting Information S1). As the excretion and mortality of zooplankton serves as a pathway for POM export to the ocean interior (Hauck et al., 2013), such a decrease in grazing proportion would lead to a redistribution of geochemical tracers and a reorganization of deep-sea carbon inventory (Figure S7 in Supporting Information S1).

Consequently, our sensitivity experiment obtains an overall carbon loss from the ocean, including the Atlantic, Southern Ocean, as well as most of the Pacific, with a significant negative DIC anomaly over 15 mmol/m^3 primarily in the upper 800 m (blue shading in Figure 6). Despite minor carbon accumulation in the intermediate Pacific, the model yields a substantial increase in atmospheric pCO_2 of 16 ppm, which is comparable to the effect of aeolian dust input (Gottschalk et al., 2019). Notably, we observe a shoaling of the e-folding depth (the depth at which 63% of sinking carbon is respired) by up to 10 m, particularly in the tropical Pacific and the subtropical gyres (Table S1 in Supporting Information S1). This is consistent with previous modeling effort suggesting a global deepening of e-folding depth by 24 m in tandem with an atmospheric pCO_2 decrease by 10–27 ppm (Kwon et al., 2009).

Instead of making realistic estimations for past carbon cycling, this simulation merely indicates a great potential of varying trophic structure in influencing oceanic carbon sequestration. With a reduced grazing proportion related to a shoaling in remineralization depth, the respired nutrients would be shifted upward and recycled more quickly (Kwon et al., 2009), and the pool of remineralized PO_4 would shrink. Consequently, the respired carbon pool would be further decreased, resulting in an additional increase in atmospheric pCO_2 (Ito & Follows, 2005; Marinov et al., 2006).

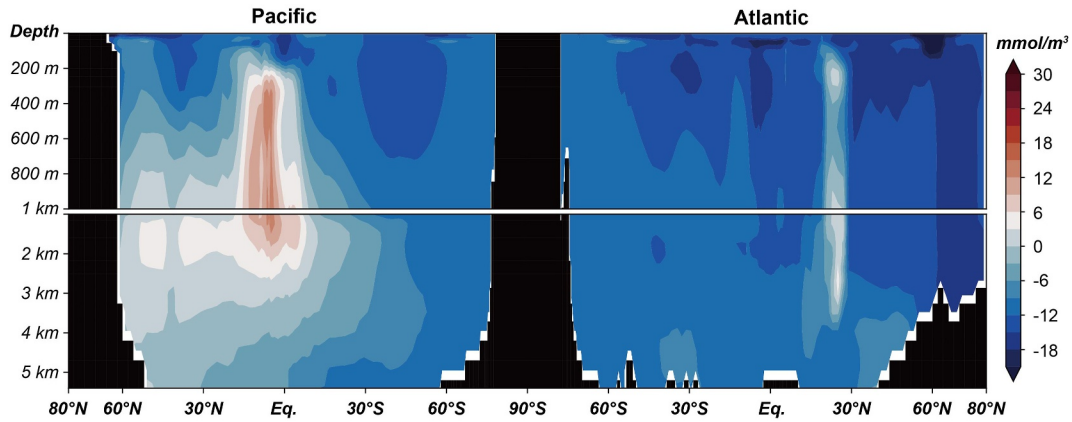


Figure 6. Changes in the basin-scale zonal mean DIC concentrations driven by a reduced grazing proportion. From the left to the right, the transect runs from the North Pacific to the Southern Ocean, and then to the North Atlantic to show the interglacial DIC anomaly (compared to initial interglacial state), with the global grazing proportion reduced by 50% while maintaining other boundary conditions. Figure shows the results of adjusted state (grazing proportion = 0.2) minus initial state (grazing proportion = 0.4). Prominent negative DIC anomalies (i.e., carbon loss) are found in the global oceans, increasing atmospheric pCO_2 by 16 ppm. Model settings following the work by Du and Tian (2025).

Our sensitivity experiment implies that the effects of dynamic trophic structures on the carbon cycle are particularly important in the mid- to low-latitude Pacific (Figure S6 in Supporting Information S1). In support, zooplankton-sourced POC accounts for more than three-quarters of total carbon export and sequestration in the modern WEP (Nowicki et al., 2022). Correspondingly, while modern tropical oceans exhibit the lowest POC export efficiency (ratio of exported to total primary production in the surface ocean) in most ecological models (e.g., DeVries & Weber, 2017; Siegel et al., 2014), they likely archive the highest transfer efficiency (higher fraction of exported organic matter that reaches the deep ocean; Henson et al., 2012), possibly due to the refractory nature of settling POC. Taken together, despite the potentially complex and nonlinear interactions between grazing proportion, BCP efficiency and carbon cycle, this study nevertheless emphasizes the need to incorporate trophic structure into future data analyses and model simulations of BCP.

5. Conclusions and Implications

This study presents the first downcore $\delta^{15}N$ records of sedimentary AAs in the western equatorial Pacific, with the attempt to quantify changes in the grazing proportion and related effects on the carbon cycle over the last 140 ka. There is good preservation of sedimentary $\delta^{15}N_{AA}$ signatures at Site MD10-3340. While downcore decreasing THAA contents and low THAA yields indicate intensive organic matter remineralization after burial, both DI and $\sum V$ suggest that degradation of AA has minimal influence on their $\delta^{15}N$. Both $\delta^{15}N_{THAA}$ and $\delta^{15}N_{Phe}$ mainly track past variations in original upper ocean nitrate $\delta^{15}N$. Moreover, the remarkable similarities between our $\delta^{15}N_{AA}$ and $\delta^{15}N_{bulk}$ records of three WEP sites further support the applicability of $\delta^{15}N_{bulk}$ in oligotrophic regions.

Our $\delta^{15}N_{AA}$ records reveal pronounced reorganization in the WEP trophic structure over the last 140 kyr. $\delta^{15}N$ gradients between the trophic and source AAs suggest a precession-dominated oscillations in the TP and thus grazing proportion in the WEP, with slightly lower grazing proportion in interglacial period compared to the Last Glacial of MIS 2–4. This orbital-scale reorganization of the planktonic trophic structure is mainly attributed to the orbital-driven changes in upper WEP stratification, with El Niño-like mean states during precession maxima leading to a smaller grazing proportion. Our sensitivity experiment with ocean-biogeochemical model further suggests that a 50% reduction in global grazing proportion during interglacial could effectively reduce carbon storage in the intermediate waters of mid- to low-latitude Pacific, substantially contributing to the rise in atmospheric CO_2.

This work underscores the large-amplitude changes in marine grazing proportion on orbital time-scale in the past, and hence proposes the necessity to incorporate variables concerning planktonic trophic structure, as a key factor determining the BCP efficiency, in the biogeochemistry-enabled ocean-atmosphere coupled modeling efforts.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Data sets are publicly accessible for the organic content, amino acid $\delta^{15}N$ data and grazing proportion calculation of Site MD10-3340 at https://doi.org/10.5281/zenodo.18043245 (Li et al., 2025a), and for carbon cycle model outputs at https://doi.org/10.5281/zenodo.18043404 (Li et al., 2025b).

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